ME 4503: Mechanics of Machine

Assignment II

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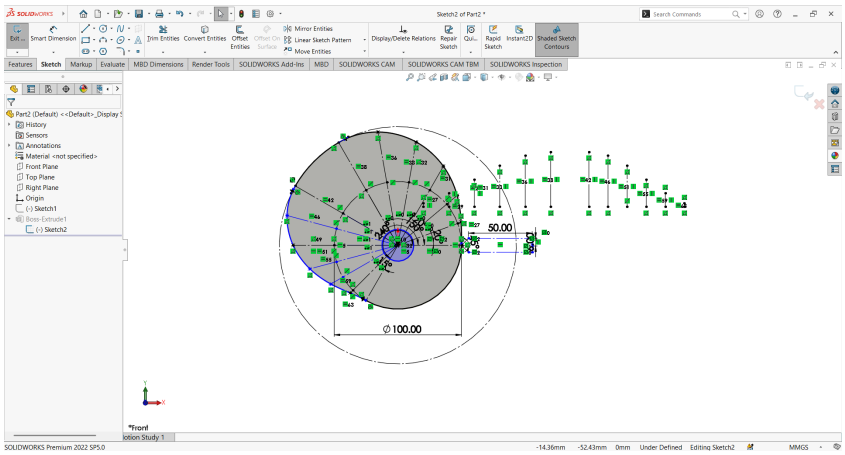
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1 Question 1: Cam Profile Design

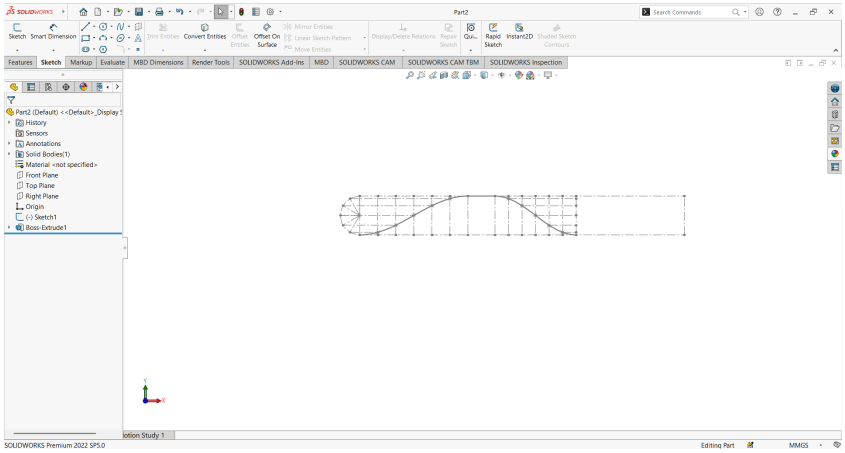
Here, we have to design a ***CAM profile*** for Knife Edge follower with the CAM specification of ***50 mm minimum radius, 25 mm shaft diameter, 230 rpm clockwise rotation***, and the follower specification of ***43 mm stroke length, SHM (Simple Harmonic Motion) during raise & fall, 43 mm rise during*** 120° ***rotation,*** 30° ***dwell, 43 mm lower during next*** 90°, ***and*** 120° ***dwell at end***.

1.1 Task

(a) Below, a screenshot clearly showing the ***minimum diameter of the CAM*** is given:



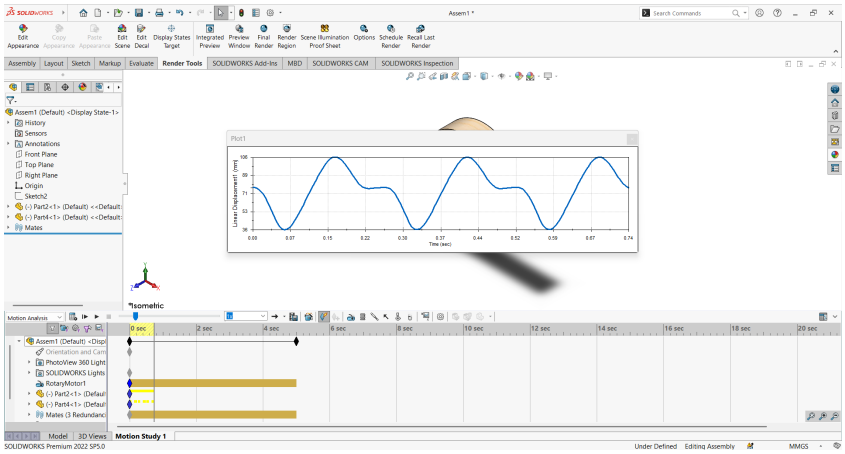
(b) Below, a screenshot of the ***displacement diagram*** is given:



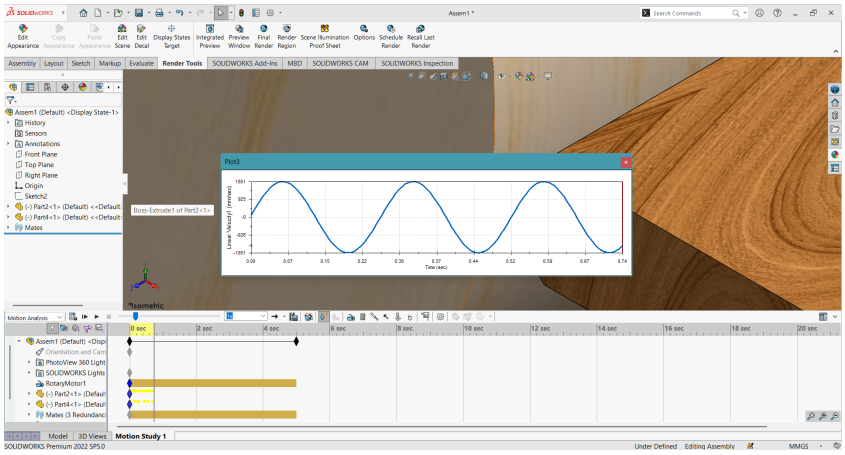
(c) Below, a screenshot of the ***complete CAM follower system*** is given:



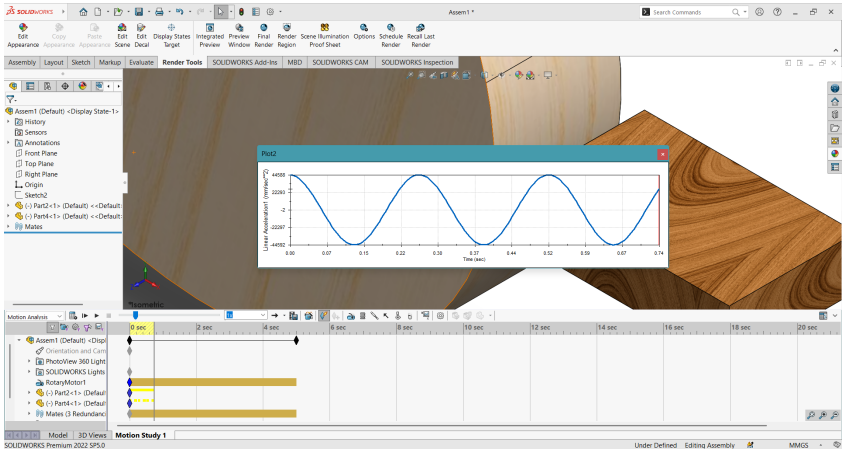
(d) Below, a screenshot of the ***Linear Displacement vs. Time Diagram*** is given:



(e) Below, a screenshot of the ***Linear Velocity vs. Time Diagram*** is given:



(f) Below, a screenshot of the ***Linear Acceleration vs. Time Diagram*** is given:



2 Question 2: Compound Gear Train Design

Here, we will design a compound gear train that reduces the rotational speed between an input shaft and output shaft. The ***input shaft*** rotates at ***1830 rpm*** in the clockwise direction. The ***output shaft*** rotates at ***180 rpm*** in the counterclockwise direction. We are considering the following constraints:

(a) *No gear in the transmission may act as an idler gear.*

(b) *The gear ratio at any individual mesh must not exceed 3:1.*

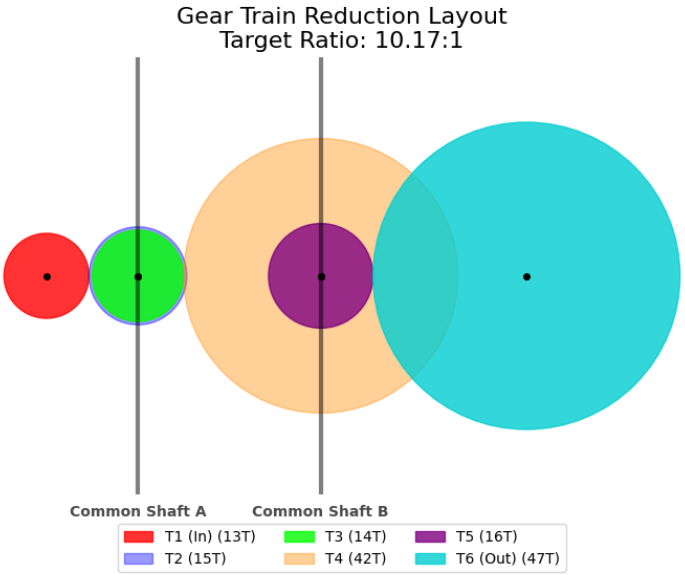
(c) *Available gears have integer tooth numbers ranging from 13 to 85.*

(d) *Only one gear exists for each tooth number, meaning the same tooth number cannot be used more than once in the system.*

(e) *If the difference between the required ratio and calculated ratio is smaller than 0.01, then the gear combination can be acceptable.*

2.1 Task

(a) Below, a ***labeled sketch*** of the gear train configuration is provided:



(b) The ***python code*** used to search for possible solutions:

```
import pandas as pd

input_rpm = 1830
output_rpm = 180
target_ratio = input_rpm / output_rpm

min_teeth = 13
max_teeth = 85
max_mesh_ratio = 3.0
tolerance = 0.01

def solve_gear_train():
    solutions = []

    for t1 in range(min_teeth, max_teeth + 1):
        for t2 in range(min_teeth, max_teeth + 1):
            r1 = t2 / t1
```

```
            if r1 > max_mesh_ratio or t1 == t2: continue

            for t3 in range(min_teeth, max_teeth + 1):
                for t4 in range(min_teeth, max_teeth + 1):
                    for t5 in range(min_teeth, max_teeth + 1):
                        for t6 in range(min_teeth, max_teeth + 1):
                            used = {t1, t2, t3, t4, t5, t6}
                            if len(used) < 6: continue

                            r2 = t4 / t3
                            r3 = t6 / t5
                            if r2 > max_mesh_ratio or r3 > max_mesh_ratio: continue

                            calc_ratio = r1 * r2 * r3
                            if abs(calc_ratio - target_ratio) < tolerance:
                                solutions.append([t1, t2, t3, t4, t5, t6, round(
calc_ratio, 4)])
                                if len(solutions) >= 10:
                                    return solutions
    return solutions

columns = ['T1 (In)', 'T2', 'T3', 'T4', 'T5', 'T6 (Out)', 'Calculated Ratio']
results = solve_gear_train()
df = pd.DataFrame(results, columns=columns)

print("\nTop 10 Candidate Gear Combinations:")
print(df.to_string(index=False))
```

(c) A ***table*** of ten candidate gear combinations produced by the program:

T1 *(In)*	***T2***	***T3***	***T4***	***T5***	***T6*** *(Out)*	***Calculated Ratio***
13	15	14	42	16	47	10.1683
13	15	16	47	14	42	10.1683
13	15	16	47	17	51	10.1683
13	15	16	47	18	54	10.1683
13	15	16	47	19	57	10.1683
13	15	16	47	20	60	10.1683
13	15	16	47	21	63	10.1683
13	15	16	47	22	66	10.1683
13	15	16	47	23	69	10.1683
13	15	16	47	24	72	10.1683

(d) The selected optimal gear set is the first one: ***T1(In) = 13, T2 = 15, T3 = 14, T4 = 42, T5 = 16 & T6 (Out) = 47***. The reasons for being chosen the first combination is:

(1) A compound gear train with three stages (three meshes) reverses the direction of rotation an odd number of times.

(a) *Stage 1: CW → CCW Stage*

(b) *Stage 2: CCW → CW Stage*

(c) *Stage 3: CW → CCW Stage*

Since the requirement is for the output shaft to rotate counterclockwise, a 3-stage design is necessary and correct.

(2) The first set is superior to others for two reasons:

(a) The maximum gear size used is ***47 teeth*** where the later design used gears up to ***72 teeth***. Smaller gears result in a more compact gearbox, lower weight, and lower material cost.

(b) Using gear counts like ***13, 47, and 17*** is generally preferred in mechanical design. These are often prime or have fewer common factors, which ensures that the same teeth don't meet as frequently.

(3) The ratios for this set are:

(a) $R_1 = 15/13 \approx 1.15$

(b) $R_2 = 42/14 \approx 3.00$

(c) $R_3 = 47/16 \approx 2.94$

All individual meshes are ≤ 3.0, satisfying constraint ***(b)***. While R_1 is quite small, keeping the largest gear to 47 teeth (instead of 72) is a significant advantage for the overall footprint of the assembly.

(4) All gears are part of a mesh that contributes to the ratio. So, there are *no idlers*.

(5) All tooth numbers $\{13, 15, 14, 42, 16, 47\}$ are unique, satisfying constraint ***(d)***.

(6) All values are within the $[13, 85]$ range.